Model Test Coverage Report - COMP3211 PIM

Group: 89. **Revision:** independent review, 2026-10-23. **Delivery:** current candidate files identified by the manifest. Application fixes and tests are included; the old source commit is not the complete delivery.

1. Reproduce the full suite

From the submitted `03_Implementation` folder in PowerShell:

```
$env:PYTHONPATH = "$PWD\src"
python -m unittest discover -s tests -t . -v
python tools/model_coverage.py
```

The independent revision passed **51 test methods, zero failures and errors**, on Python **3.11.9, 3.12.3 and 3.13.5**. These comprise 25 model, 13 controller, 7 storage, 4 real-process acceptance and 2 measurement-tool tests. `subTest` cases are included within method counts. All original 45 methods remain; this review added four application tests and two tool tests.

Host: Windows 11, build 26200, confirmed using the operating-system record. Python 3.11's `platform.release()` reports Windows 10 on this host; that interpreter label in raw logs does not identify a second machine. The 3.11.9 runtime is an official standalone embeddable distribution configured for normal imports. A separate **51-test run with `-S` (site packages disabled)** also passed. No third-party runtime dependency is required or bundled.

Raw logs are `independent-verification-2026-10-23-python311.txt`, `independent-verification-2026-10-23-python312.txt` and `independent-verification-2026-10-23-python313.txt`. Coverage uses the same suffixes after `independent-model-coverage-2026-10-23-`. Repository logs are in `docs/reports/`; candidate copies are in `03_Implementation/evidence/`.

2. Model source-line coverage

The full 51-test suite achieved **416/421 model source lines (98.81%)** on each of the three interpreters. The numerator and denominator include only `src/model/*.py`. Controller, storage, tests and tools are not included as measured source. CLI subprocess execution is outside the parent trace session.

Model module	Covered / executable source lines	Coverage
`model/__init__.py`	0/0	n/a
`model/errors.py`	4/4	100.00%
`model/manager.py`	95/95	100.00%
`model/records.py`	57/57	100.00%
`model/search.py`	226/231	97.84%
`model/validation.py`	34/34	100.00%
Total model	**416/421**	**98.81%**

The tool uses standard-library `trace` to collect executed lines and its private `_find_executable_linenos` helper to identify candidate executable lines. It now excludes nonpositive/noninteger entries (`0` and `None`), which refer to interpreter bookkeeping, not source lines. It prints excluded-entry counts and actual missing source lines for each module. This private CPython helper is version-dependent, so results name the measured interpreter and are not a promise for future releases.

The prior Python 3.13 report counted 416/430 (96.74%); nine denominator entries were `0` or `None`. Removing these produces 416/421 without changing the numerator or model code. On Python 3.11 six such entries were removed from 427. The increase is **a corrected measurement, not newly covered model behavior**. The remaining five real lines, `search.py` 205-209, belong to the unused `_Parser._accept` helper. They remain in the denominator; dead code was not deleted or artificially called to inflate coverage.

Two fixture-based regressions first failed against the old tool and passed after correction: a docstring-only module has 0/0 executable source lines, and an uncalled real function body remains in the denominator and is listed as missing. Nested tracing restores the previous tracer after each fixture run.

3. Model unit tests alone

To distinguish model unit evidence from the full suite, a separate Python 3.11.9 run executed only the **25 model unit methods**, using the same coverage algorithm. It covered **415/421 lines (98.57%)**. The extra missing line is `manager.py` 22, the invalid record-type branch exercised through the boundary tests in the full suite. Other module counts are unchanged.

Reproduce this narrower check from the same folder:

```
$check = "from tools import model_coverage as c; c.TESTS /= 'model'; "
$check += "raise SystemExit(c.main())"
python -c $check
```

The resulting log is `independent-model-units-only-2026-10-23.txt`. Its runner prints the default tool command in the header; the command above is the actual narrower invocation. The full-suite percentage must not be described as coverage produced only by model unit tests.

4. Assertion quality and limits

Existing assertions cover all four PIR types, nine mutable fields, five text-search fields, three time fields and all three time comparisons, invalid inputs, immutable replacement, stable IDs, complete snapshot validation and failure atomicity. This review added an independent truth-table oracle for six three-atom Boolean compositions over all eight assignments (48 subcases), rather than assertions derived from the parser's own output.

Two fixes have observed failing-then-passing regressions: deeply nested malformed JSON is converted to a FORMAT error with the old state and next ID preserved, and a line-buffered terminal receives each prompt before input blocks. The nesting failure was reproduced on Python 3.11; the same sample takes a different rejection path on newer interpreters. A 40-import AST audit and the `-S` test run support the standard-library restriction.

This is source-line coverage, not branch, condition, mutation or exhaustive requirements coverage. No course minimum coverage percentage is specified. NFR-01's exact network-disabled/no-graphical-session environment remains untested. Human recordings, genuine identities and the official declaration remain separate deliverables. The final candidate's clean-extraction log records verification of the actual ZIP contents.